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Machine Safety Specialist (CE)

Tab 1 – Module 5: Machinery Regulation

TAB1: principles machine safety
Tab1-5/5 20231010

Attention: This module is not part of the exam

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Tab1-5/5-Sh.2

Machinery Directive becomes 'Machinery Regulation'

- The current Machinery Directive is being revised and replaced by the new Machinery Regulation.
- Regulation 2023/1230
- Application date 20-01-2027
- Until then, there is a transition period during which manufacturers can prepare themselves.

29.6.2023 EN Official Journal of the European Union L 165/1

I

(Legislative acts)

REGULATIONS

REGULATION (EU) 2023/1230 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL
of 14 June 2023
on machinery and repealing Directive 2006/42/EC of the European Parliament and of the Council and Council Directive 73/361/EEC
(Text with EEA relevance)

THE EUROPEAN PARLIAMENT AND THE COUNCIL OF THE EUROPEAN UNION,

Having regard to the Treaty on the Functioning of the European Union, and in particular Article 114 thereof,

Having regard to the proposal from the European Commission,

After transmission of the draft legislative act to the national parliaments,

Having regard to the opinion of the European Economic and Social Committee ([1]),

Acting in accordance with the ordinary legislative procedure ([2]),

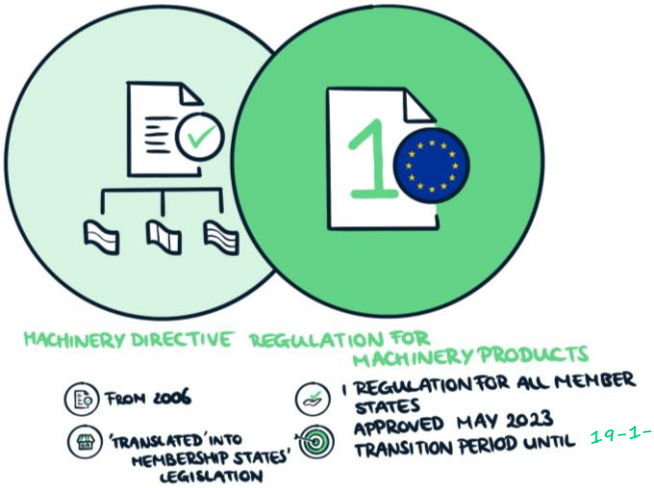
2

Tab1-5/5-Sh.3

Machinery Directive becomes 'Machinery Regulation'

Why a Regulation instead of a Directive?

- *Transposition into national legislation*
- *Differences in interpretation*
- *No Recitals*



Tab1-5/5-Sh.4

Entry into force and application

- The regulation is applied starting 42 months after the regulation's entry into force.
- The regulation enters into force on the 20th day following its publication in the Official Journal of the European Union.
- This sets the application date at 20-01-2027.
- From that moment, it will no longer be permitted to 'CE mark' a machine based on the Machinery Directive.
- A 'hard transition moment' has been chosen. Earlier application of the regulation is not permitted [*1].

*[*1] It is possible to fully comply with the requirements of the Machinery Regulation and to communicate this accordingly, but the declaration will remain based on the Machinery Directive until the application date of the regulation.*

Entry into force and application

- So we have time to prepare…

Machinery Regulation – Recital (3)

"Experience with the application of Directive 2006/42/EC has shown inadequacies and inconsistencies in the product coverage and conformity assessment procedures. It is therefore necessary to ***improve, simplify and adapt the provisions set out in that Directive to the needs of the market*** *and provide clear rules in relation to the framework within which products within the scope of this Regulation may be made available on the market."*

Machinery Regulation – Recital (12)

*"Recently, more **advanced machinery**, which is less dependent on human operators, has been introduced on the market. Such machinery is working on defined tasks and in structured environments, yet it can **learn to perform new actions** in this context and become **more autonomous**. Further refinements to machinery, already in place or to be expected, include real-time processing of information, problem solving, mobility, sensor systems, learning, adaptability, and capability of operating in unstructured environments (for example construction sites). The Commission Report on the safety and liability implications of Artificial Intelligence, the Internet of Things and Robotics of 19 February 2020, states that the emergence of new digital technologies, like artificial intelligence, the Internet of things and robotics, raises new challenges in terms of product safety. The report concludes that the current product safety legislation, including Directive 2006/42/EC, contains a number of gaps in this respect that need to be addressed. **Thus, this Regulation should cover the safety risks stemming from new digital technologies**."*

Machinery Regulation – Recital (19)

*"The evolution of the machinery sector has resulted in the growing use of digital means and **software plays a more and more important role in machinery design**. Consequently, the definition of machinery should be adapted. In this respect, machinery missing only the upload of software intended for the specific application foreseen by the manufacturer, and which is the subject of the conformity assessment procedure of the machinery, should fall under the definition of machinery and not under the definitions of related products or partly completed machinery. Furthermore, the definition of safety components should cover not only physical devices but also digital devices. In order to take into account the increasing use of software as a safety component, **software that performs a safety function and which is placed independently on the market should be considered a safety component**.*

Machinery Regulation – Recital (25)

*"**Other risks related to new digital technologies are those provoked by malicious third parties that have an impact on the safety of products** within the scope of this Regulation. In this respect, **manufacturers should be required to adopt proportionate measures which are limited to the protection of the safety of the product** within the scope of this Regulation. This does not preclude the application to products within the scope of this Regulation of other Union legal acts specifically addressing cybersecurity aspects."*

Machinery Regulation – Definitions (Article 3)

New definitions have been added, amongst others (not limitative):

- Source code
- Professional user
- Lifetime
- Recall
- Withdrawal
- Economic operator
- Distributor
- Safety function

"source code": means the currently installed version of the software of a product within the scope of this Regulation, written in a programming language so that it is unambiguous and understandable to humans;

Impact for machine manufacturer

- The general objective of the regulation has not changed compared to the Machinery Directive.
- Products falling within the scope must be freely tradable within the European Economic Area when their safety level is ensured.
- Therefore, the process to be followed by the machine builder will not be significantly different.
- Technically, there has mainly been an update to the 'state of the art,' which is reflected in aspects such as 'Machine Learning,' 'Cyber Security,' and 'safety-related software.'

Impact for machine manufacturer

There still needs to be…

- … carried out a conformity assessment (the machine must be (demonstrably) in compliance with the applicable essential requirements);
- …prepared and signed a declaration of conformity;
- … applied a CE-marking;
- … provided instructions for use.

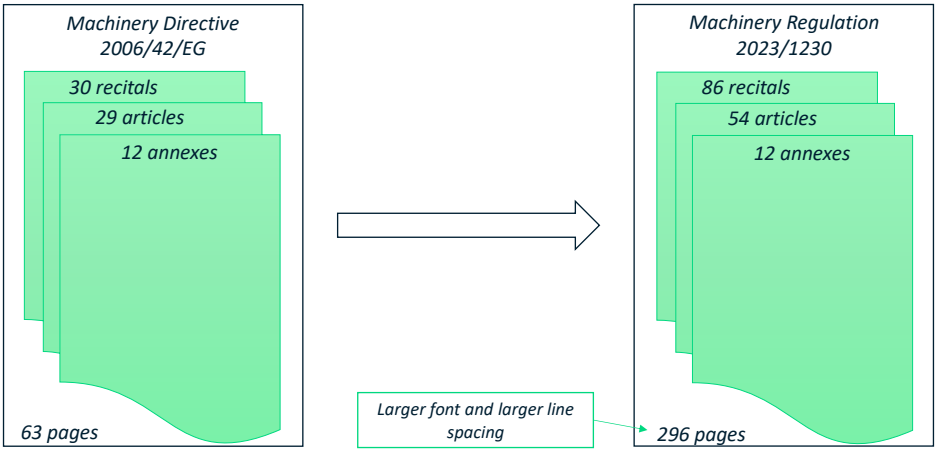
For many machine manufacturers, the technical impact of the regulation will likely be limited. The question is whether new essential safety and health requirements apply and what is necessary to demonstrably bring the machine into compliance with them.

Impact for the machine user

- The impact of the regulation on the party purchasing a new machine will also be limited.
- They will still receive a machine with CE marking, a Declaration of Conformity, and instructions for use.
- Typically, the machine manufacturer will likely only provide digital instructions for use (although they are still obligated to provide a 'paper version' upon request from the user).
- Apart from the Machinery Regulation, it is increasingly important during a procurement process to determine whether new/application-specific cybersecurity requirements need to be met concerning a cyber-resilient integration of the machine into the end user's OT system (refer to Jeroen Heijkants' lecture on this subject).

Lay-out of the Regulation is changed

- "Those familiar with the Machinery Directive may sometimes need to search for the relevant sections in the Machinery Regulation. However, the structure itself has not been changed:"



Tab1-5/5-Sh.15

Lay-out of the Regulation is changed

- Recasting articles from Machinery Directive to Machinery Regulation:

Machinerichtlijn 2006/42/EG	Machineverordening 2021/0105
Artikel 1	Artikel 2
Artikel 2	Artikel 3
Artikel 3	Artikel 8 en 9
Artikel 4	-
Artikel 5	Artikel 7
Artikel 6	Artikel 4
Artikel 7	Artikel 17
Artikel 8	Artikel 45
Artikel 9	-
Artikel 10	Artikel 42

Machinerichtlijn 2006/42/EG	Machineverordening 2021/0105
Artikel 11	Artikel 41 en 44
Artikel 12	Artikel 21
Artikel 13	Artikel 22
Artikel 14	Artikel 24 en 40
Artikel 15	Artikel 23
Artikel 16	Artikel 19
Artikel 17	Artikel 20
Artikel 18	Artikel 47
Artikel 19	-
Artikel 20	-

Machinerichtlijn 2006/42/EG	Machineverordening 2021/0105
Artikel 21	Artikel 45 en 51
Artikel 22	Artikel 46
Artikel 23	Artikel 48
Artikel 24	-
Artikel 25	Artikel 49
Artikel 26	-
Artikel 27	-
Artikel 28	Artikel 52
Artikel 29	Artikel 52

Tab1-5/5-Sh.16

Lay-out of the Regulation is changed

- Recasting annexes from Machinery Directive to Machinery Regulation:

Onderwerp van de bijlage	Machinerichtlijn 2006/42/EG	Machineverordening 2021/0105
Essentiele veiligheids- en gezond-heidseisen	I	III
Verklaring van overeenstemming	II	V
CE-Markering	III	Artikel 19 verwijst naar EU/765/2008
Categorieën gevaarlijke machines (Hoog risico)	IV	I (Deel A en deel B)
Veiligheidscomponenten (indicatief)	V	II
Montagehandleiding voor niet-vol-tooide machines	VI	X
Technisch Dossier	VII (A en B)	IV (A en B)
Beoordeling van de overeenstemming met interne controle van de bouw van een machine	VIII	VI
EG-Typeonderzoek	IX	VII
Volledige kwaliteitsborging	X	IX[1]
Eenheidskeuring	-	VIII
Minimumcriteria voor aanmelding van keurende instanties	XI	Artikel 28

Machinery regulation – some differences

('Annex I~~X~~ machines')

- The Regulation includes an overview in Annex I of 'dangerous machinery,' which is divided into:
  - Part A: Dangerous machinery requiring involvement of a Notified Body (NoBo) (EC type-examination, unit verification, or full quality assurance).
  - Part B: Dangerous machinery where the manufacturer may assess conformity themselves, provided harmonized (type-C) standards are applied (Internal production control).

- The European Commission must periodically review the classification of dangerous machinery between Annex I Part A and Part B, where:
  - Machines may be moved between A ⇄ B.
  - Machines may be added to Part A or Part B.
  - Machines may be removed from Part A or Part B.

**Two years after the regulation comes into force, an initial assessment will be conducted, followed by subsequent reviews every five years.*

Machinery regulation – some differences

('Annex I~~X~~ machines')

Bijlage I – Categorieën machines of machineproducten waarop een van de in artikel 21, lid 2 en 2a, bedoelde procedures toegepast moet worden		
Nr.	**Deel A** (procedures zoals beschreven in artikel 21 (2) zijn van toepassing):	**Deel B** (procedures zoals beschreven in artikel 21 (2a) zijn van toepassing):
1		Cirkelzagen (eenbladig en meerbladig), voor de bewerking van hout en materialen met gelijkaardige fysieke eigenschappen of voor de bewerking van vlees en materiaal met gelijkaardige fysieke eigenschappen. *[zie de verordening voor een complete opsomming]*
2		Vlakschaafmachines met handmatige toevoer voor houtbewerking.
3		Eenzijdige schaafmachines met geïntegreerde voeding, met handmatige toevoer en/of afvoer voor houtbewerking.
4		Lintzagen van het volgende type: Met handmatige toevoer en/of afvoer voor de bewerking van hout en materialen met gelijkaardige fysieke eigenschappen. Of voor de bewerking van vlees en materialen met gelijkaardige fysieke eigenschappen. *[zie de verordening voor een complete opsomming]*
5		Gecombineerde machines van de in de punten 1 tot en met 4 en 7 bedoelde typen voor de bewerking van hout en materialen met gelijkaardige fysieke eigenschappen.
6		Pennenbanken met verschillende spillen met handmatige toevoer voor houtbewerking.
7		Freesmachines met verticale as, met handmatige toevoer, voor de bewerking van hout en materialen met gelijkaardige fysieke eigenschappen.
8		Draagbare kettingzaagmachines voor houtbewerking.
9		Persen, met inbegrip van buigmachines, voor koude metaalbewerking, waarbij het materiaal met de hand wordt toegevoerd en/of afgevoerd en de beweegbare werktuigen een slaglengte kunnen hebben van meer dan 6 mm en een snelheid van meer dan 30 mm/s.
10		Machines voor het spuitgieten en persen van kunststoffen met handmatige toevoer of afvoer van het materiaal.
11		Machines voor het spuitgieten en persen van rubber met handmatige toevoer of afvoer van het materiaal.
12		Machines voor ondergrondse werkzaamheden van de volgende typen. *[zie de verordening voor een complete opsomming]*
13		Met de hand geladen vuilniswagens met perssysteem.

Bijlage I – Categorieën machines of machineproducten waarop een van de in artikel 21, lid 2 en 2a, bedoelde procedures toegepast moet worden		
Nr.	**Deel A** (procedures zoals beschreven in artikel 21 (2) zijn van toepassing):	**Deel B** (procedures zoals beschreven in artikel 21 (2a) zijn van toepassing):
14	Verwijderbare mechanische transmissiesystemen, inclusief hun afschermingen.	
15	Afschermingen voor verwijderbare mechanische transmissiesystemen.	
16	Hefbruggen voor voertuigen.	
17		Hijs- en hefwerktuigen voor het heffen van personen of van personen en goederen waarbij een gevaar voor een vrije val van meer dan drie meter bestaat.
18	Draagbare bevestigingswerktuigen met explosieve lading en andere slagwerktuigen.	
19		Beveiligingsinrichtingen voor de detectie van personen.
20		Aangedreven beweegbare afschermingen met vergrendeling voor de beveiliging van de in de in punten 9 (deel A), 10 en 11 (deel B) bedoelde machines.
21		Logische eenheden ter verzekering van veiligheidsfuncties.
22		Kantelbeveiligingsinrichtingen (ROPS).
23		Constructies ter bescherming tegen vallende voorwerpen (FOPS).
24	Veiligheidscomponenten met volledig of gedeeltelijk zelflerend gedrag (AI) gebruikmakend van machine-learning, bedoeld om in veiligheidsfuncties toegepast te worden.	
25	Embedded systems voor machines met volledig of gedeeltelijk zelflerend gedrag (AI) gebruikmakend van machine-learning, bedoeld om in veiligheidsfuncties toegepast te worden maar die niet afzonderlijk op de markt zijn gebracht.	

After application of the machine regulation a NoBo must always be involved for these machines.

Added

Machinery regulation – some differences

(Essential health and safety requirements)

- Now in Annex III of the Machinery Regulation.
- The structure (a general part and several additional parts) and the subjects to which requirements are applied remain largely unchanged.
- Regarding certain topics such as machine learning, cybersecurity, and (safety) software, additional requirements have been added.

Machinery regulation – some differences

(Essential health and safety requirements)

Some examples of added or modified requirements:

[1.1.2] *…specific functional test procedures, essential to enable it to be tested, adjusted, maintained and used safely.*

[1.1.9] *The machinery or related product shall be designed and constructed so that the connection to it of another device, via any feature of the connected device itself or via any remote device that communicates with the machinery or related product does not lead to a hazardous situation.*

[1.2.1] *the tracing log of the data generated in relation to an intervention and of the versions of safety software uploaded after the machinery or related product has been placed on the market or put into service is enabled for five years after such upload, exclusively to demonstrate the conformity of the machinery or related product with this Annex further to a reasoned request from a competent national authority.*

[1.6.2] *In the case of machinery or related products into which persons shall enter for operation, adjustment, maintenance or cleaning, the machinery accesses shall be dimensioned and adapted for the use of rescue equipment in such a way that an emergency rescue of the persons is possible.*

[Etc.]

Machinery regulation – some differences

(Substantial modification)

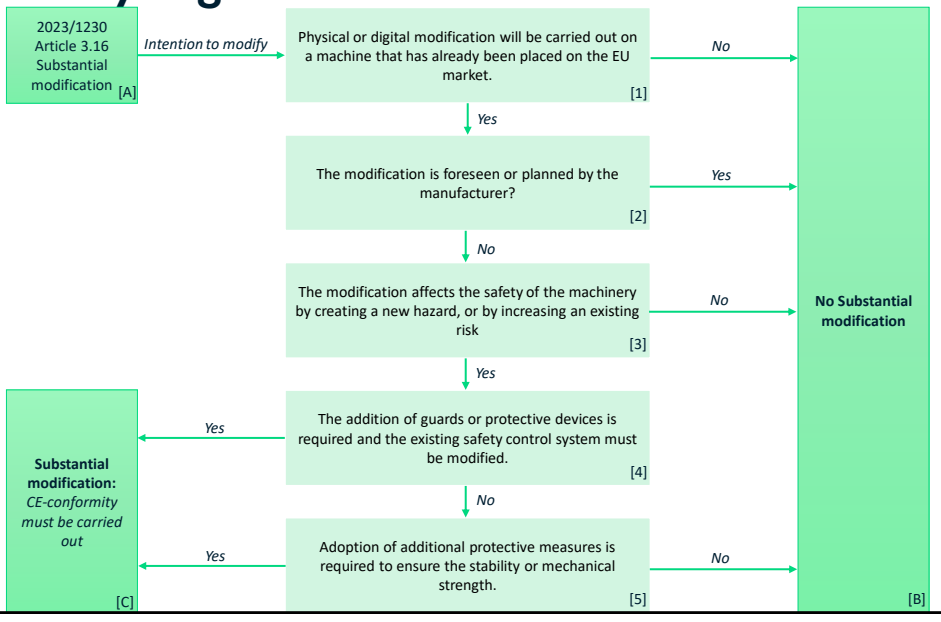
- in Article 3 (nr 16) the following definition is given:

"Substantial modification": a modification of machinery or a related product, by physical or digital means after that machinery or related product has been placed on the market or put into service, which is not foreseen or planned by the manufacturer, and which affects the safety of that machinery or related product, by creating a new hazard, or by increasing an existing risk, which requires:

a) the addition of guards or protective devices to that machinery or related product the processing of which necessitates the modification of the existing safety control system; or

b) the adoption of additional protective measures to ensure the stability or mechanical strength of that machinery or related product

Machinery regulation – some differences

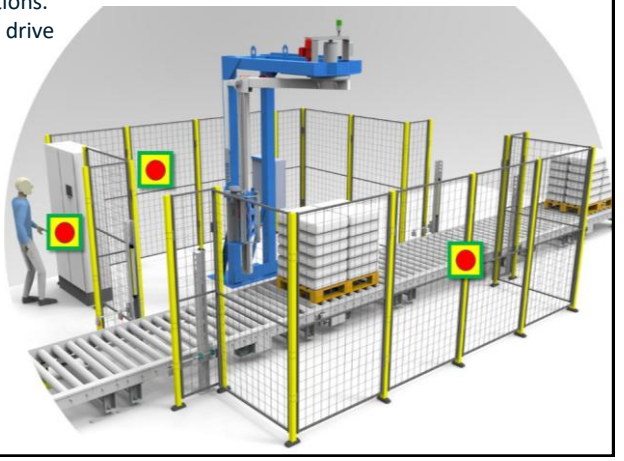


Machinery regulation – some differences

(Substantial modification)

Example 1: adding of additional emergency stop buttons to a pallet wrapping machine

- A manufacturing company wants to equip a pallet wrapper with additional emergency stop buttons at several locations. Pressing these emergency stop buttons should stop the drive of the pallet wrapper.
- The technical department has assessed that the emergency stop buttons can be hardware-connected to the existing emergency stop circuit of the machine.



Machinery regulation – some differences

(Substantial modification)

Example 2: increasing the wrapping speed of a pallet wrapper

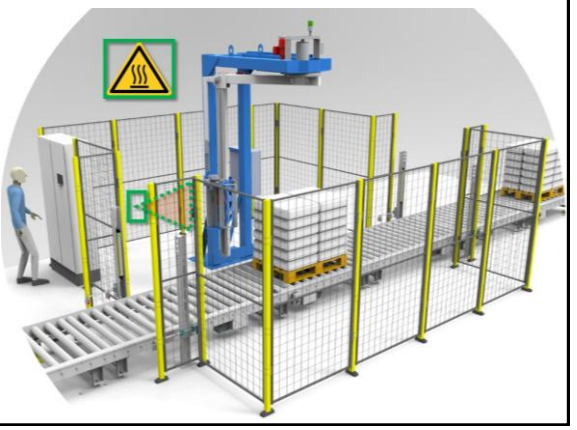
- A manufacturing company wants to increase the production capacity of an existing pallet wrapper by 20%.
- The wrapping speed of the pallet wrapper will be increased by adjusting the gear transmissions.
- The technical department has assessed that this should not pose any issues concerning the control system. However, points).dat it has been determined that the current structure anchoring are likely not capable of withstanding the altered dynamic forces. Additional measures will therefore be taken, such as reinforcing certain critical parts of the structure (enlarging and strengthening the base and adding extra braces and anchoring points).

Machinery regulation – some differences

(Substantial modification)

Example 3: adding a film heating system

- A manufacturing company is going to modify a pallet wrapper to use a special type of film. For this film, it is essential that it is heated during the wrapping process.
- The technical department has assessed that this should not pose any issues concerning the control system. However, it has been determined that several new hazards are introduced (accessible hot parts, new risk of entrapment).
- Additional measures:
  - Installing perforated guards over the heating element
  - Implementing the heating system as a type of blower. Blowing hot air onto the film from a distance (to prevent entrapment)
  - Adding a pictogram to warn about hot parts.



Machinery regulation – some differences

(Substantial modification)

Is there a substantial modification in these 3 examples?

- Who thinks that example 1 (emergency stop buttons) is a Substantial modification?
- Who thinks that example 2 (increasing speed) is a Substantial modification?
- Who thinks that example 3 (film heating) is a Substantial modification?

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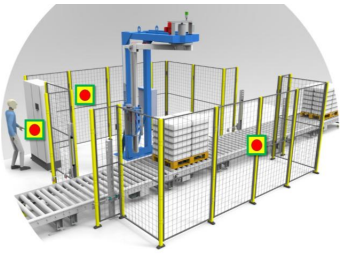
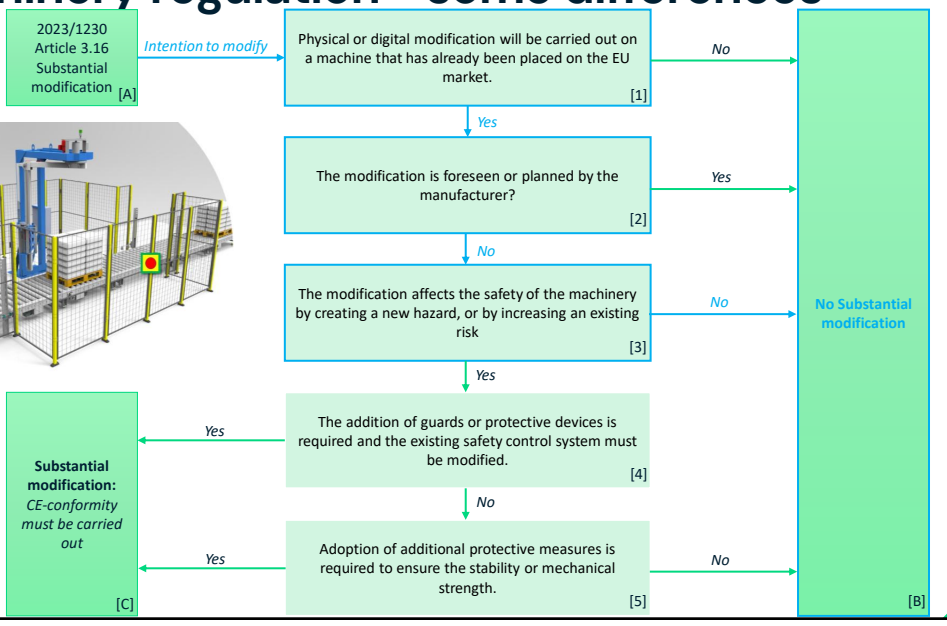
Machinery regulation – some differences





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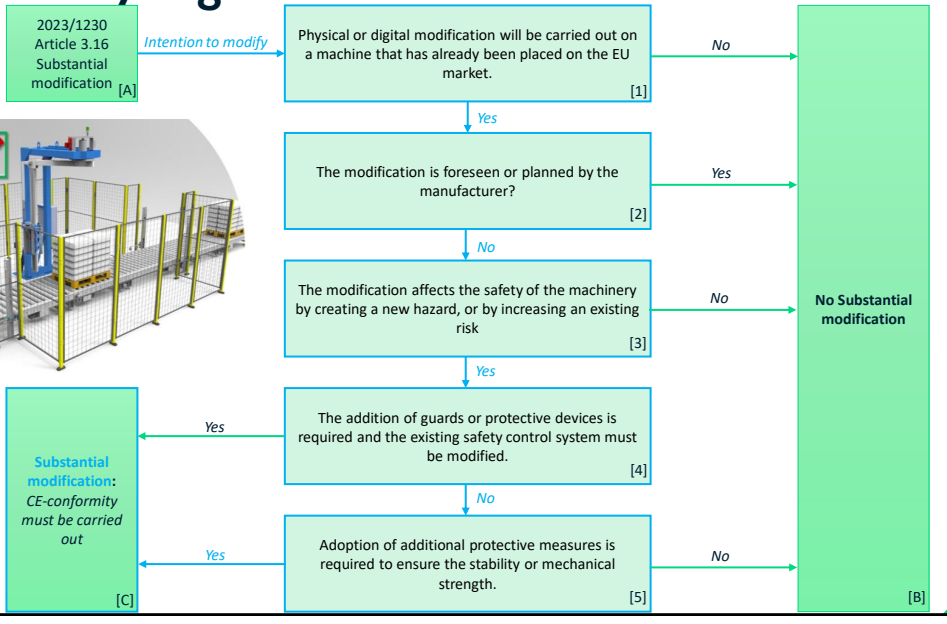
Machinery regulation – some differences



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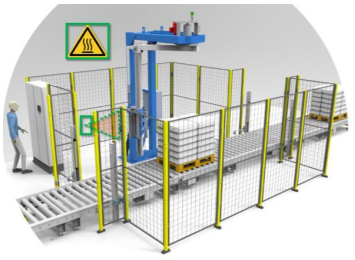
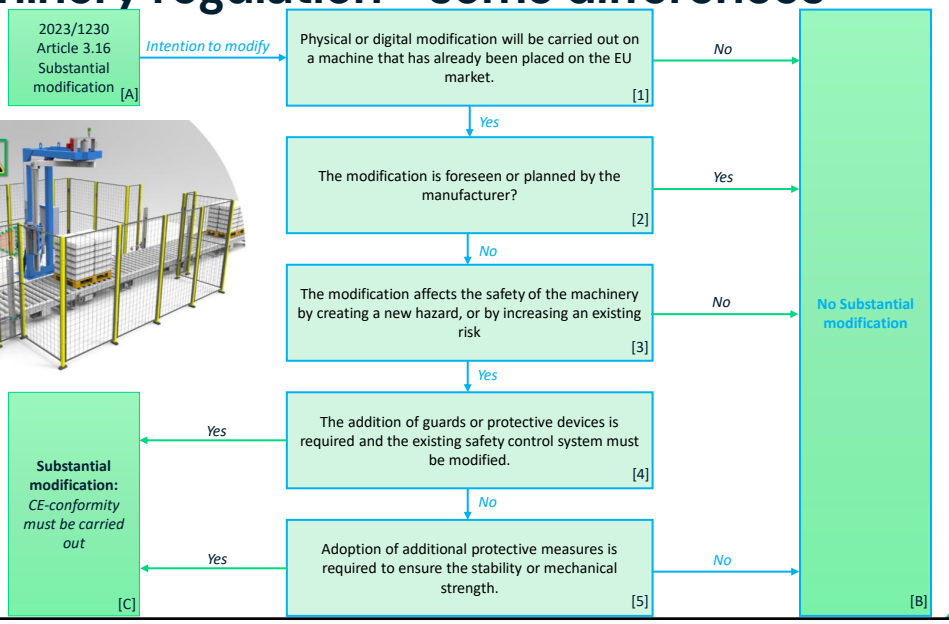
Machinery regulation – some differences





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Machinery regulation – some differences

(Substantial modification)

Some principles regarding A Non-Substantial modification summarized:

- Implementing risk-reducing measures based on a conducted Risk Inventory & Evaluation (RI&E) on an existing machine will generally not result in a Substantial Modification.
- This also means that merely modifying the safety control system or adding guards and protective measures to an existing machine (based on an RI&E) does not need to be considered a Substantial Modification, provided that the modification does not introduce new hazards or increase existing risks.
- When the assessment concludes that a change does not need to be considered a Substantial Modification, the aspect of (machine) safety must still be carefully considered. The only difference is the legal framework within which the safety of the modified machine must be evaluated (Occupational Health and Safety Act instead of the Machinery Regulation).

Machinery regulation - some differences
(Substantial modification)

Some principles a Substantial modification summarized:

- For a Substantial Modification, a conformity assessment must be carried out on the entire modified machine in accordance with Article 15 of the regulation.

- Recital 26 only states the following:

"…[…] To avoid unnecessary and disproportionate burdens, the person making the Substantial Modification should not be required to repeat tests and create new documentation for machinery or related products that are part of an assembly of machinery and are not affected by the modification."

- This aligns with the work instruction "Assessing Modified Machines" from the Dutch Labour Inspectorate (https://www.nlarbeidsinspectie.nl/publicaties/richtlijnen/2022/05/23/werkinstructie-beoordelen-van-gewijzigde-machines)

- After completion, state in the declaration of conformity that it concerns a substantially modified machine, and also specify what the original machine was.

Machine regulation – Key differences
(economic operators)

A couple of new recitals and definitions are added in relation with the following economic operators. These have been aligned with the new Market Surveillance Regulation (2019/1020):

- Importer (Article 12)

- Distributor (Article 13)

Machine regulation – Key differences

(Instructions for use)

- According the machine regulation it is allowed that the Instructions for use are delivered in a digital format
- The manufacturer shall provide a paper format of the instructions for use at the request of the user at the time of purchase (free of charge within one month after request).
- For machines, intended for non-professional users (consumers) or machines that can, under reasonably foreseeable conditions, be used by non-professional users, the manufacturer shall:
  - provide in paper format the safety information that is essential for putting the machinery or related product into service and for using it in a safe way
  - other parts can be provided in a digital format

Machine regulation – Key differences

(Instructions for use)

- When the instructions for use are provided in digital format, the manufacturer shall:

  a) mark on the machinery or related product, or, where that is not possible, on its packaging or in an accompanying document, how to access the digital instructions;

  b) present them in a format that makes it possible for the user to print and download the instructions for use and save them on an electronic device;

  c) make them accessible online during the expected lifetime of the machinery and for at least 10 years after the placing on the market of the machinery.

Machine regulation – Key differences

(Instructions for use)

- In article 10 of the regulation it is stated that the instructions for use shall be in a language which can be easily understood by users, as determined by the Member State concerned.
- The requirement for delivering the instructions for use also in the original language is not in the machine regulation.

Machine regulation – Key differences

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